ME448 - Thermal Systems Design
~10 min. Lab Project Presentation:

Ice Rink Design & Analysis

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Reason for Proposal



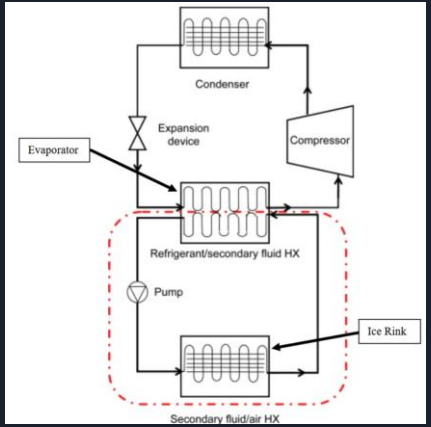
Key reasons:

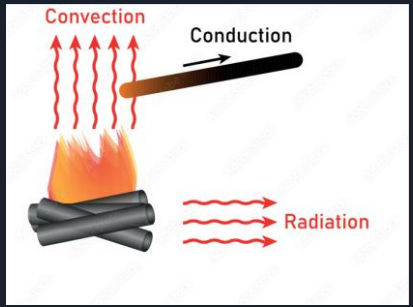
- Dr. Pascual wants to bring back a ice rink similar to what Cal Poly had in the 1960s
- Assigned class project
- Non-trivial design problem with low & high level considerations
- Learning exercise in analysis & design

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Background

- Lots of EES issues……
- **Complexity of system:**
  - Multi-mode heat transfer analysis
  - Fluid dynamics (analysis & design)
  - Thermal systems design
  - Properties being dependent on results (iterative solving)
  - Multiple start up time cases
  - Component selection
- **Heat transfer modeling:**
  - 2D conduction (FEA)
  - Phase change in ice
  - Radiation exchange on ice surface
  - Convection heat loss on ice surface
  - Conduction heat loss through insulation into the ground
  - Heat flux between brine and ice
  - Effects form the tent over the ice
  - Thermal resistances of ice, concrete, and insulation
  - Transient (2 cases) and steady state analysis

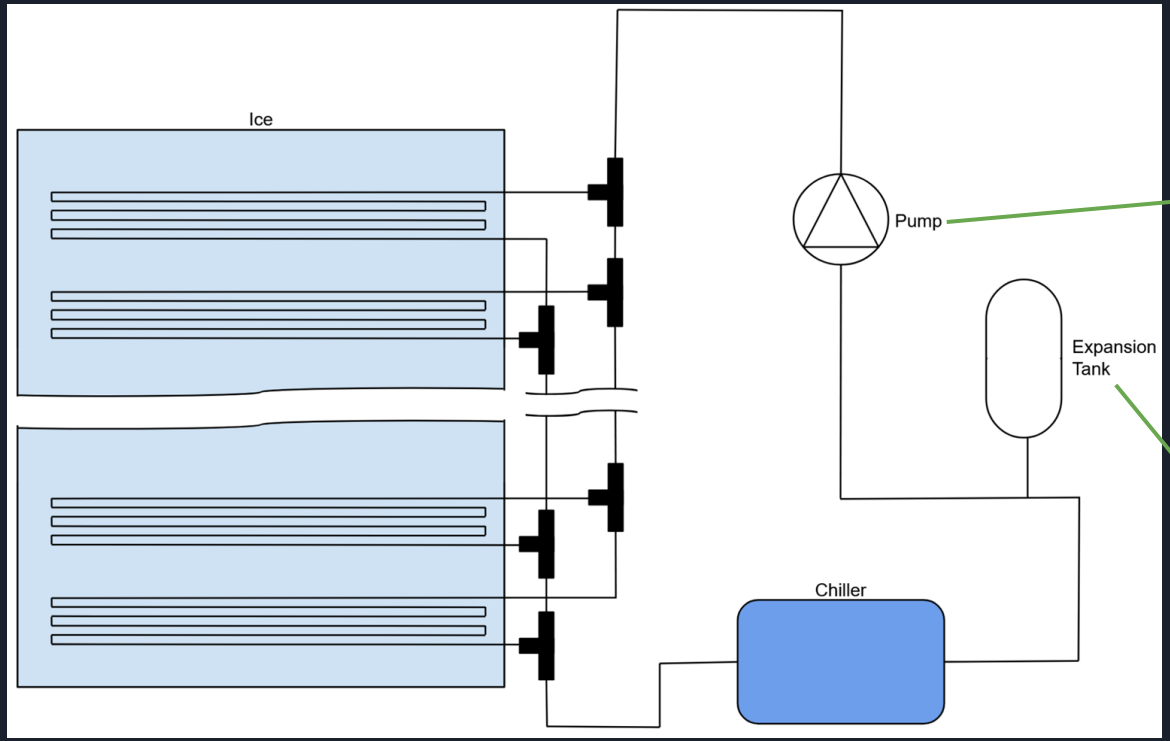


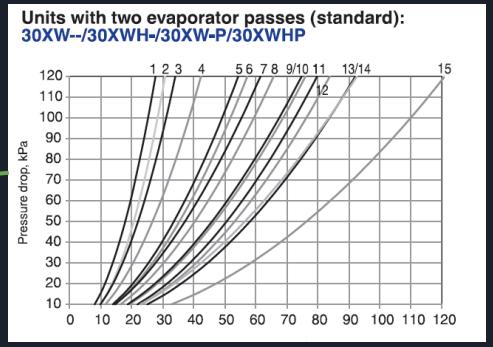


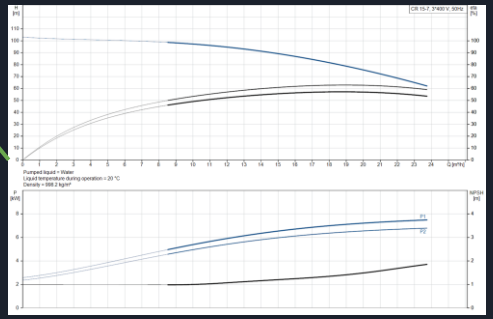
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Basic System Design (high level)

This short presentation is abridged and has left out many key details, for a full description see our report.







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Analysis Techniques

EES:

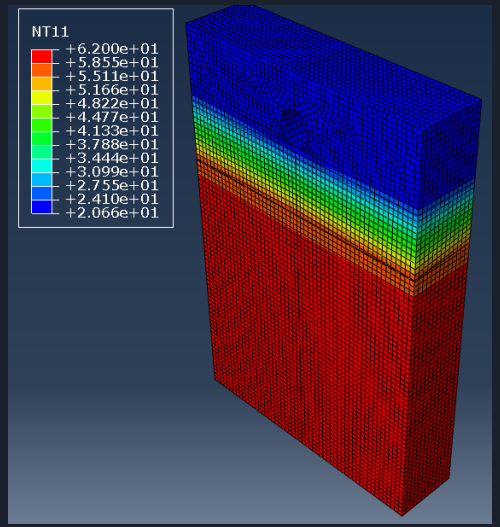


- Total calculations divided into 3 scripts
- Scripts go over:
  - Heat load calculations
  - Thermal resistance
  - Heat transfer
  - Flow and pressure drop
  - Pump and system design
  - Chiller and expansion tank sizing
- Scripts work by solving for design parameters as systems of equations

ABAQUS:

- Symmetry based model with full output
- Assess effects of tube spacing
- Estimate the relevant thermal resistances of ice and the insulation
- Assess heat flux into and out of the system

FEA results showing nodal temperature distribution:



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Economic & Design Considerations
(why we did what we did)

- Tube sizing & spacing
- Pump Selection
  - 8 hour
  - 24 hour
- Expansion valve selection
  - 8 hour
  - 24 hour

Expansion Tank Calculations

Given: $P_{\text{fill,abs}} = 29.7$ psi, $P_{\text{max,abs}} = 54.7$ psi.

8-hour Scenario

$$\Delta V = V_{\text{system}} \times \left(\frac{1}{\rho_{\text{min}}} - \frac{1}{\rho_{\text{max}}} \right) \times \rho_{\text{max}}$$

$$\Delta V_{8hr} = 66.84\,\text{ft}^3 \times \left(\frac{1}{1188} - \frac{1}{1192} \right) \times 1192 \approx 0.225\,\text{ft}^3$$

$$V_{\text{tank,min}} = \frac{0.225\,\text{ft}^3}{\left(\frac{54.7}{29.7} - 1\right)} \approx 0.321\,\text{ft}^3$$

$$V_{\text{tank,selected}} = 0.321\,\text{ft}^3 \times 1.2 \approx 0.385\,\text{ft}^3$$

24-hour Scenario

$$\Delta V = V_{\text{system}} \times \left(\frac{1}{\rho_{\text{min}}} - \frac{1}{\rho_{\text{max}}} \right) \times \rho_{\text{max}}$$

$$\Delta V_{24hr} = 66.84\,\text{ft}^3 \times \left(\frac{1}{1187} - \frac{1}{1190} \right) \times 1190 \approx 0.169\,\text{ft}^3$$

$$V_{\text{tank,min}} = \frac{0.169\,\text{ft}^3}{\left(\frac{54.7}{29.7} - 1\right)} \approx 0.241\,\text{ft}^3$$

$$V_{\text{tank,selected}} = 0.241\,\text{ft}^3 \times 1.2 \approx 0.289\,\text{ft}^3$$

Selected Expansion Tanks

- **8-hour Scenario**: Amtrol THERM-X-TROL ST-5 ($\approx$0.4 ft^3)
- **24-hour Scenario**: Wessels FXA-100 ($\approx$0.3 ft^3)

Both tanks handle the required expansion, pressures up to 125 psig, and are compatible with $CaCl_2$ brine.

What Sets Our Project Apart

- FEA in ABAQUS instead of Excel (far better mesh)
- Sceptical approach to cost estimation
- Follow through with (unrequired) assessment of level of uncertainty in presented values at RSmeans
- Sceptical approach to heat transfer modeling (including research on heat transfer through permeable membranes)
- Willingness to install tubing connectors for ~$40/connector when the time comes to buld
- Identification of strange errors and lack of supporting evidence for table 1 (Optimum Fluid Velocity in Pipes)
- Likely the most set apart & unique project submitted (therefore most useful)

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Conclusion

- System design balancing efficiency, functionality, cost, and feasibility for 8 hour and 24 hour startup scenarios using iterative and analytical modeling techniques.
- Heat transfer analysis to partly optimize startup and steady-state power requirements, highlighting energy trade-offs between speed and energy efficiency.
- Fluid dynamic analysis and modeling to select pumps, piping/tubing dimensions, and flow rates to ensure efficient brine circulation and adequate pressure management.
- Chiller and expansion tank selection was done to meet cooling load requirements and maintain system stability during operation.

Recommended future work:

- Validate theoretical results with physical testing, especially for the chiller and pump.
- Improve efficiency with adaptive control systems and explore more advanced control methods.
- Analyze the overall structure for better design.
- Assess electrical needs.
- Design system with setup, removal, and storage efficiency considerations in mind.
- Optimize insulation materials for performance.

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Questions?

